

Statistics Project Report:
Influence of GDP per Capita on Cancer Death Rates

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Abstract:

This study investigates the statistical relationship between a country's economic status, measured by Gross Domestic Product (GDP) per capita, and its cancer death rate for the year 2019. A cross-sectional analysis was performed using 2019 data for 167 countries from the data platform Kaggle, and websites Our World Data as well as countryeconomy.com. Key variables, Cancer Death Rate per 100,000 people and GDP per capita, were calculated. The GDP per capita data had a log transformation applied to it when being analyzed due to its original nature of being highly right-skewed. The relationship was analyzed using the Pearson Correlation test and a Log-Linear Regression model. The analysis revealed moderate, positive, and highly significant correlation between GDPs per capita and cancer death rate. The log-linear regression model was also highly significant with the model being able to explain 46.9% of the variance in cancer death rates. Therefore, the original proposition where greater national wealth is associated with lower cancer death rates is disproven. Additionally, confounding factors that were not included in the model also influenced its accuracy in the relationship between GDPs per Capita and cancer death rates. The most conspicuous factor includes increased life expectancy, differences in lifestyle, and cause of death reporting accuracy in wealthier nations.

Introduction:

According to the National Cancer Institute (NCI), "Cancer is a disease in which some of the body's cells grow uncontrollably and spread to other parts of the body." (National Cancer Institute, 2021, para. 1) The origin can vary within the human body, and survival rates vary based on cancer type. The core characteristics of cancer are uncontrollable cell growth, ability to traverse to any other location within the human body and cause genetic mutations to the DNA of the cells that control cell growth and cell division. An individual can obtain cancer for various reasons which include but are not limited to family genome, environmental exposures, lifestyle factors, infections, and age.

To ensure current cancer prevention knowledge is expanded upon, this study aims to explore how a country's Gross Domestic Product (GDP) per capita impacts its cancer death rates. It is hypothesized that there is a significant inverse statistical relationship between the cancer death rate per 100,000 people and a country's GDP per capita.

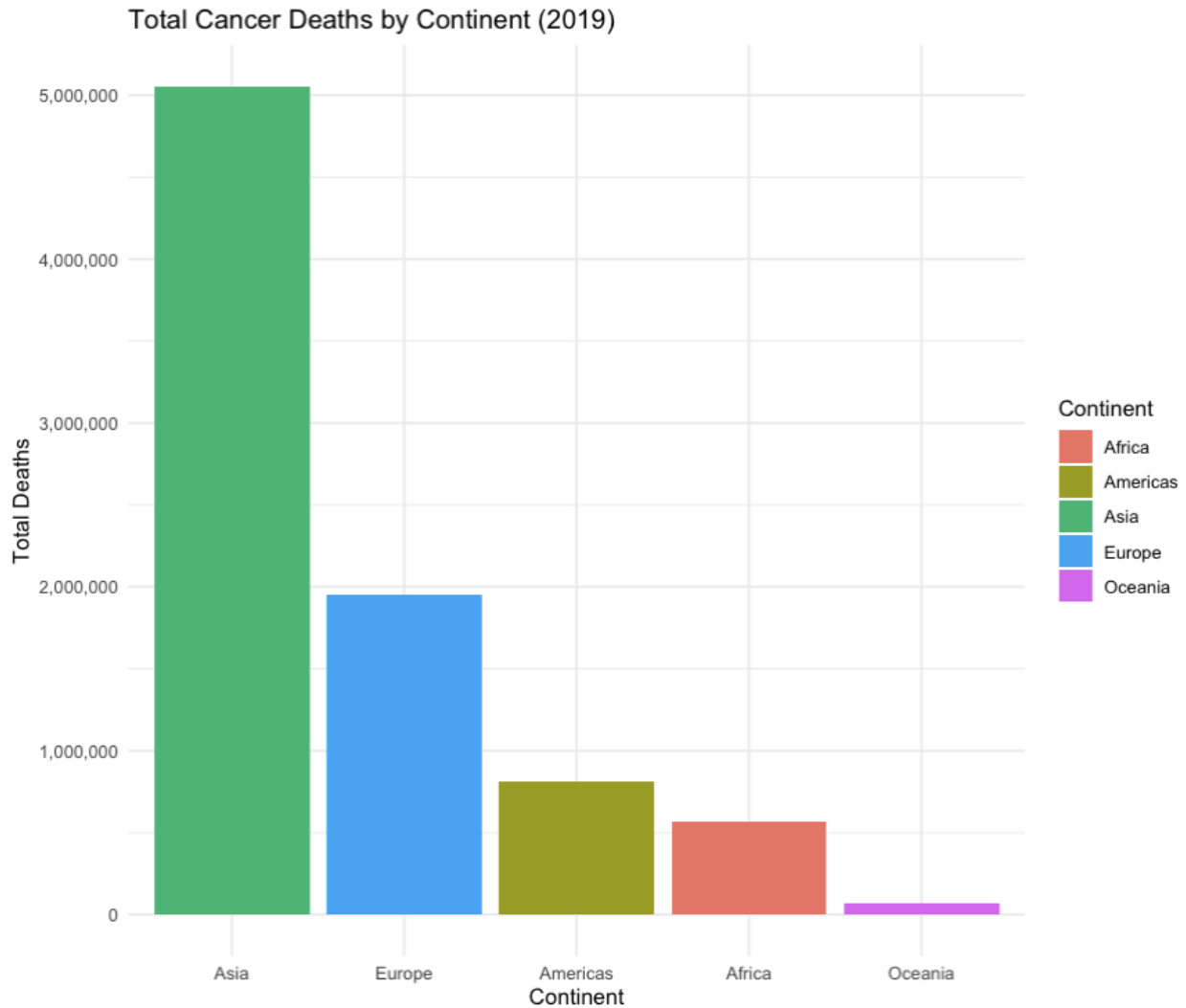
Method:

The data for this analysis was sourced from the “Cancer” dataset (hrterhrter, 2021), a data repository hosted on the Kaggle platform. This repository is a snapshot of the comprehensive “Cancer” dataset originally compiled and published by “Our World in Data” (Rose & Ritchie, 2015).

The “Our World in Data” team collects this data from primary global health organizations, with the most recognizable being the Institute for Health Metrics and Evaluation (IHME) Global Burden of Disease (GBD) study. According to the source, the IHME methodology involves aggregating and standardizing data from a wide variety of sources, including national registries and hospital records. For regions where this data is not ideal, such as missing information, the IHME employs statistical modeling to generate comprehensive global estimates, proving to be accurate.

From the “Cancer” collection, the “total-cancer-deaths-by-type” dataset was selected. The analysis was performed using the programming language R. The data was filtered to include only the year 2019. For each country, all distinct cancer death types were summed to create a “Total_Deaths” variable. This disregarded both sex and age of the patient's death.

Figure 1: Total Cancer Deaths by Continent (2019)



The following plot displays the cancer death rate distribution based on continent. Please note that this plot does not take into consideration for both sex and age of the patient.

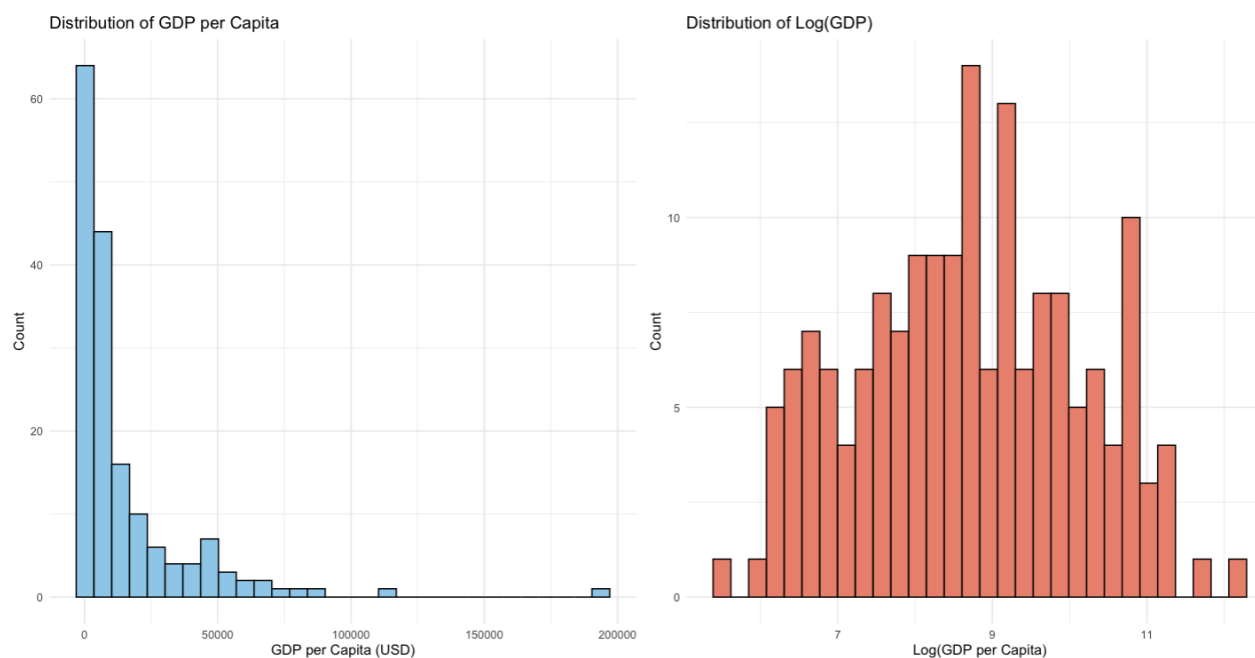
This cancer data was then merged with 2019 GDP in millions sourced from “countryeconomy.com” and population size from “populationpyramid.net”. To ensure data consistency, several cleaning steps were taken:

- Non-country entities (e.g., “World,” “Asia,” “High-income countries”) were filtered out.
- Country names were standardized across the datasets (e.g., “Russian Federation” was changed to “Russia”).
- Any rows with missing or zero values for GDP or Population were removed, resulting in a final analysis dataset for 167 countries

The two key metrics that were calculated for each country were the “GDP_per_Capita” and the “Death_Rate_per_100k”. More specifically, the “GDP_per_Capita” variable was calculated by multiplying the total GDP by 1 million and dividing it by the country's population. The GDP is multiplied by 1 million because the data recorded on “countryeconomy.com” is in millions. The “Death_Rate_per_100k” is calculated by dividing the total number of deaths by the population and then multiplying it by one hundred thousand.

After calculating the “GDP_per_Capita” variable, the data was noticeably highly right-skewed. To ensure that the assumptions of linear regression were met, a natural log transformation was applied to this variable. Using this newly modified data, a Pearson correlation test was conducted on the untransformed “GDP_per_Capita” and the “Death_Rate_per_100k” variables. Subsequently, a log-linear regression model was fitted to test the hypothesis where the log of the “GDP_per_Capita” is the predictor variable, and the “Death_Rate_per_100k” is the response variable.

Figure 2: GDP per Capita Transformation



As shown above, the original GDP per Capita was not normally distributed. As a result, a log transformation was applied to ensure a normal distribution.

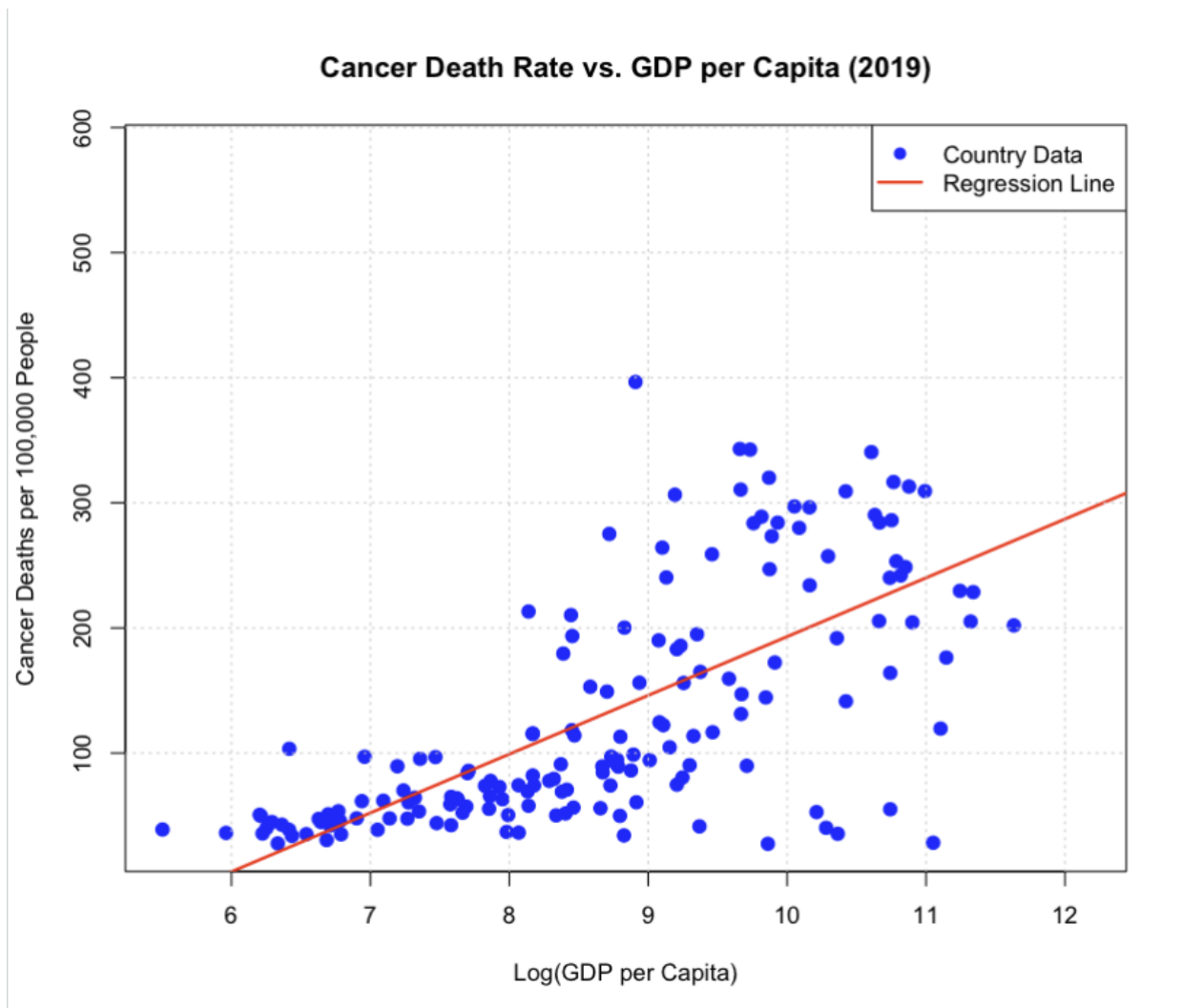
It is important to note that this analysis might be affected by several key confounding variables such as the age distribution of the population, lifestyle of a country's population, or differences in healthcare system quality and cancer screening program. As a result, please keep in mind that all these listed variables might have an influence on the observed relationship within this study.

Results:

The statistical analysis yielded several key findings. The Pearson's Correlation Test between the untransformed variables revealed a moderate, positive, and highly statistically significant correlation between "GDP_per_Capita" and "Death_Rate_per_100k" variables. More specifically, the correlation coefficient was approximately 0.597, the Confidence Interval ranged from 0.490 to 0.687, and the p value was less than $2.2 \cdot 10^{-16}$. With the p value being drastically smaller than the chosen significance level of 0.05 or 5%. Since the p value is less than the significance level, the null hypothesis that there is no correlation between the GDP and cancer deaths is false. For this reason, it can be concluded that countries with higher GDP per capita tend to have higher cancer death rates.

The log linear regression model was also highly statistically significant. The f test yielded a ratio of 145.4 which means that the model can explain the variation in the cancer death rate 145.4 times stronger than unexplained variation out of the 165 degrees of freedom from the 167 countries. The p value for the model was less than $2.2 \cdot 10^{-16}$ which is also extremely less than the significance level of 0.05 or 5%. With the p value being less than the significance level, the null hypothesis that asserts there is no relationship between the GDP and cancer deaths, is proven false, indicating a significant relationship exists. The model also produced an Adjusted R Squared value of 0.4653, which suggests that the logarithm of GDP per capita accounts for approximately 46.5% of the variance in cancer deaths per one hundred thousand people. The intercept coefficient was -276.594 with a p value of less than $1.52 \cdot 10^{-13}$ and the slope coefficient was 46.97 with a p value of less than $2 \cdot 10^{-16}$. The positive slope coefficient confirms the positive relationship between the "GDP_per_Capita" and "Death_Rate_Per_100k" where the cancer death rate is expected to increase by approximately 47 deaths per one hundred thousand people.

Figure 3: Scatter Plot of Cancer Death Rate vs. Log(GDP per capita)



As shown in figure 3, the scatter plot of the data and the fitted red regression line visually confirm the positive relationship. The blue data points are widely dispersed but clearly follow the overall upward trend of the regression line.

Discussion:

The results of this analysis can potentially be misleading. The positive correlation of 0.597 between national wealth (GDP per capita) and cancer death rates contradicts the common assumption that higher income pays for better healthcare and in turn lead to better health outcomes. However, this is not the case due to the confounding variable age. More specifically, higher GDP per capita is strongly linked to better public health, sanitation, and

medical care, which leads to a longer average life expectancy within a country. Since cancer is primarily a disease of aging, countries where people live longer will have a higher overall rate of cancer deaths due to its larger elderly population. Typically, wealthier nations use more advanced healthcare technology. As a result, doctors can accurately diagnose and record cancer as the official cause of death. In lower GDP countries, people are more susceptible to other health risks that can lead to death, prior to the development of cancer. With limited medical access, a death from cancer can potentially be misattributed to other causes or go unrecorded.

These findings do not imply that wealth causes cancer, or that healthcare is ineffective. Instead, they suggest that the relationship between the wealthiness of a country and cancer death rates consist of a more complex relationship, influenced by demographic and lifestyle transitions that accompany economic development. The predictive model produced an Adjusted R Squared value of 0.465 which indicates even though it is a significant predictor, more than half of the variation in cancer death rates is explained by other factors, such as the two explained above.

Conclusion:

This study found a significant, moderate, positive correlation between a country's GDP per capita and its cancer death rate per 100,000 people for the year 2019. The log linear regression model confirmed this with the Adjusted R Squared value confirming that approximately only 46.5% of the variance in death rates could be accounted for. The main highlight is that a simple, direct relationship does not exist between both variables. The positive association with this assumption is likely due to the longer life expectancy in wealthier nations which results in a larger elderly population that is more susceptible to cancer, as well as differences in lifestyle risks and reporting accuracy.

The most important improvement that should be taken into consideration is to analyze this relationship again but instead using age standardized cancer death rates. This would control the differences in population age structures between countries and provide a much clearer picture of the true underlying risks. Another way to improve the results of this report would be to include the confounding variables where a multiple regression model could be built to include other predictors. Examples include but are not limited to healthcare expenditure, smoking or obesity rates, and life expectancy, to observe if the effect of GDP persists. An

alternative research direction can be related to analyzing specific cancer types which all have very different risk factors and potential relationships with economic development.

While the 2019 data is recent, its accuracy is a known limitation due to the use of IHME modeling in some regions. However, a strong, significant result across 167 nations provides a strong foundation for future investigations.

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